Acceptance Test Report

BCIS Subscription Billing and Collection System: mandatory acceptance tests (spec section 7) and automated test evidence (spec 7.1, deliverable D5).

Generated	October 11, 2026 at 3:37 PM (Asia/Manila)
Source	commit `071dda2`
Environment	Node v24.15.0, PostgreSQL 18.6, Windows_NT 10.0.19045
Billing months used	2026-08 (the spec's "August"), 2026-09 ("September"), 2026-10
Overall result	PASS all tests passed

How these tests run

The acceptance suite (`apps/api/src/acceptance/acceptance.test.ts`) uses the spec's own test data and drives the system through its HTTP API, exactly as the desktop app does: each user signs in and sends requests with their session token, and the server checks permissions on every request. Plans, subscribers and installations are set up directly; every step being tested (billing, payments, GCash, reversals, collector batches, backups) goes over HTTP. The tests run against a dedicated test database (`bcis_test`).

To reproduce: `pnpm --filter ./apps/api acceptance` (this report), or `pnpm --filter ./apps/api test` (all API tests).

Summary

ID	Scenario	Test data	Expected result	Result	Time
AT-01	Exact payment	Invoice ₱999; payment ₱999	Remaining ₱0; invoice PAID; ledger balanced; receipt created.	PASS	249 ms
AT-02	Partial payment	Invoice ₱999; payment ₱500	Remaining ₱499; PARTIALLY_PAID; allocation and ledger correct.	PASS	44 ms
AT-03	Advance payment	Monthly ₱1,000; payment ₱3,000	Allocation/credit follows documented policy without losing value.	PASS	138 ms
AT-04	Oldest-first arrears	August 2026 ₱999 + September 2026 ₱999; payment ₱1,200	August ₱0; September ₱798 remaining.	PASS	116 ms
AT-05	Duplicate GCash reference	GCash reference 970396420405, verified once (with proof image), then submitted again	System blocks or explicitly warns according to documented verification policy.	PASS	137 ms

ID	Scenario	Test data	Expected result	Result	Time
AT-06	Payment reversal	Reverse an incorrectly posted ₱999 cash payment	Original remains visible; linked reversal exists; balances restore; actor/reason audited.	PASS	74 ms
AT-07	Collector balanced remittance	Cash collected ₱20,000; remitted ₱20,000	Difference ₱0; batch may reconcile/close.	PASS	151 ms
AT-08	Collector shortage	Cash collected ₱20,000; remitted ₱19,500	₱500 shortage displayed; cannot silently close as balanced.	PASS	149 ms
AT-09	Concurrent users	Three signed-in sessions (two cashiers, one administrator) each post 5 payments of ₱100 and read the payment list, all at the same time	No data corruption, duplicate numbering or unauthorized cross-session effect.	PASS	836 ms
AT-10	Authorization	Cashier calls admin-only user and backup endpoints directly (no UI)	Server rejects operation even if API is called directly.	PASS	26 ms
AT-11	Duplicate billing	Run billing generation and finalization for September 2026 twice more	No duplicate finalized invoice for same service account/period.	PASS	65 ms
AT-12	Backup/restore	Create a backup, post a ₱100 payment, restore the backup	Restored database passes integrity check and expected records return.	PASS	5514 ms

Results in detail

AT-01 Exact payment PASS

Test data: Invoice ₱999; payment ₱999

Expected (spec): Remaining ₱0; invoice PAID; ledger balanced; receipt created.

Observed:

- Invoice INV-000089: total ₱999.00, paid ₱999.00, remaining ₱0.00, status paid.
- Ledger: debits ₱999.00 = credits ₱999.00; balance ₱0.00.
- Receipt RCPT-000098 issued (cash, ₱999.00), all of it applied to INV-000089.

Automated test: `AT-01 Exact payment: ₱999 on a ₱999 invoice leaves ₱0, PAID, ledger balanced, receipt created`

Also covered by: `apps/api/src/payments/service.test.ts`, `packages/shared/src/allocation.test.ts`

AT-02 Partial payment PASS

Test data: Invoice ₱999; payment ₱500

Expected (spec): Remaining ₱499; PARTIALLY_PAID; allocation and ledger correct.

Observed:

- Invoice INV-000090: paid ₱500.00, remaining ₱499.00, stored status partially_paid (shown as "overdue" because the due date has passed).
- Allocation: ₱500.00 of receipt RCPT-000099 to INV-000090.
- Ledger balance ₱499.00 (debits ₱999.00, credits ₱500.00).

Automated test: `AT-02 Partial payment: ₱500 on ₱999 leaves ₱499, PARTIALLY_PAID, allocation and ledger correct`

Also covered by: `apps/api/src/payments/service.test.ts`, `packages/shared/src/allocation.test.ts`

AT-03 Advance payment PASS

Test data: Monthly ₱1,000; payment ₱3,000

Expected (spec): Allocation/credit follows documented policy without losing value.

Observed:

- Policy: a payment pays open bills oldest-first; the rest is kept as credit and applied oldest-first when the next month is finalized.
- On posting: ₱1,000.00 to INV-000091 (September 2026), ₱2,000.00 kept as credit; ledger -₱2,000.00 (in the subscriber's favour).
- After billing October 2026: INV-000104 paid from credit; receipt RCPT-000100 now ₱2,000.00 applied + ₱1,000.00 credit = ₱3,000.00. Ledger -₱1,000.00.

Automated test: `AT-03 Advance payment: ₱3,000 on a ₱1,000 plan becomes credit used by later bills, nothing lost`

Also covered by: `apps/api/src/payments/service.test.ts`, `packages/shared/src/allocation.test.ts`

AT-04 Oldest-first arrears PASS

Test data: August 2026 ₱999 + September 2026 ₱999; payment ₱1,200

Expected (spec): August ₱0; September ₱798 remaining.

Observed:

- INV-000088 (August 2026): ₱999.00 applied, remaining ₱0.00, paid.
- INV-000092 (September 2026): ₱201.00 applied, remaining ₱798.00, partially_paid.

Automated test: `AT-04 Oldest-first arrears: ₱1,200 on two ₱999 bills clears the older, leaves ₱798 on the newer`

Also covered by: `apps/api/src/payments/service.test.ts`, `packages/shared/src/allocation.test.ts`

AT-05 Duplicate GCash reference PASS

Test data: GCash reference 970396420405, verified once (with proof image), then submitted again

Expected (spec): System blocks or explicitly warns according to documented verification policy.

Observed:

- Policy: a GCash reference can be used once; the server refuses a second submission (unique index on the normalized reference).
- First submission verified: payment posted for ₱999.00.
- Second submission: HTTP 409 DUPLICATE_REFERENCE: "GCash reference 970396420405 is already used by receipt RCPT-000102 (account BCIS-001019)."
- Payments for the subscriber: 1 before and 1 after the attempt.

Automated test: `AT-05 Duplicate GCash reference: a verified reference cannot be used again`

Also covered by: `apps/api/src/payments/gcash.test.ts`, `apps/api/src/payments/routes.test.ts`, `apps/api/src/payments/schema.test.ts`

AT-06 Payment reversal PASS

Test data: Reverse an incorrectly posted ₱999 cash payment

Expected (spec): Original remains visible; linked reversal exists; balances restore; actor/reason audited.

Observed:

- Receipt RCPT-000103 is still on file with status reversed; reversal by Test administrator: "Posted to the wrong subscriber".
- Invoice INV-000094 back to ₱999.00 owing (unpaid).
- Ledger balance ₱1,998.00 before, ₱1,998.00 after; two entries added (payment credit and reversal debit), none removed.
- Audit: payment.reverse by Test administrator, reason "Posted to the wrong subscriber".

Automated test: `AT-06 Payment reversal: original stays visible, reversal linked, balance restored, actor and reason audited`

Also covered by: `apps/api/src/payments/service.test.ts`

AT-07 Collector balanced remittance PASS

Test data: Cash collected ₱20,000; remitted ₱20,000

Expected (spec): Difference ₱0; batch may reconcile/close.

Observed:

- Batch CB-000007: cash collected ₱20,000.00 (two field collections of ₱10,000), remitted ₱20,000.00.
- Reconciled: difference ₱0.00, balanced. Closed: status closed.

Automated test: `AT-07 Collector balanced remittance: ₱20,000 collected and remitted, difference ₱0, batch closes`

Also covered by: `apps/api/src/collection/reconciliation.test.ts`, `apps/api/src/collection/batch-routes.test.ts`, `packages/shared/src/collection-batches.test.ts`

AT-08 Collector shortage PASS

Test data: Cash collected ₱20,000; remitted ₱19,500

Expected (spec): ₱500 shortage displayed; cannot silently close as balanced.

Observed:

- Batch CB-000008: collected ₱20,000.00, remitted ₱19,500.00.
- Reconcile as balanced: HTTP 409 "The figures changed: the difference is now ₱500.00 shortage. Review the batch and confirm again."
- Reconcile without a reason: HTTP 400 (Explain the shortage or overage.)
- Reconciled with reason: difference -₱500.00, shortage.
- Close confirming ₱0: HTTP 409 "This batch was reconciled with a difference of ₱500.00 shortage. Confirm that figure to close it.". Close confirming the shortage: closed, shortage kept on the batch.

Automated test: `AT-08 Collector shortage: ₱20,000 collected, ₱19,500 remitted shows ₱500 short and cannot close as balanced`

Also covered by: `apps/api/src/collection/reconciliation.test.ts`, `apps/api/src/collection/batch-routes.test.ts`, `packages/shared/src/collection-batches.test.ts`

AT-09 Concurrent users PASS

Test data: Three signed-in sessions (two cashiers, one administrator) each post 5 payments of ₱100 and read the payment list, all at the same time

Expected (spec): No data corruption, duplicate numbering or unauthorized cross-session effect.

Observed:

- 30 requests in parallel, all HTTP 200. 15 receipts RCPT-000108 to RCPT-000122: no repeats, no gaps.
- Each subscriber's balance went down by exactly ₱500.00 (₱1,998.00, ₱1,998.00, ₱1,998.00 → ₱1,498.00, ₱1,498.00, ₱1,498.00).
- Each receipt names the user of the session that posted it.
- After one cashier signed out: that token gets 401; the other two sessions still get 200.
- Larger runs: apps/api/src/concurrency/at09.test.ts (billing, 30 payments, field collections, adjustments in parallel) and the three-PC script `pnpm --filter ./apps/api concurrency`.

Automated test: `AT-09 Concurrent users: three sessions posting and reading at once keep numbers and balances right`

Also covered by: `apps/api/src/concurrency/at09.test.ts`, `apps/api/src/concurrency/three-clients.ts (manual, 3 PCs)`

AT-10 Authorization PASS

Test data: Cashier calls admin-only user and backup endpoints directly (no UI)

Expected (spec): Server rejects operation even if API is called directly.

Observed:

- POST /admin/users as cashier: HTTP 403 FORBIDDEN.
- GET /admin/users as cashier: HTTP 403 FORBIDDEN.
- POST /backups as cashier: HTTP 403 FORBIDDEN.
- GET /backups as cashier: HTTP 403 FORBIDDEN.
- POST /backups without a token: HTTP 401.
- Users 5 → 5, backups 0 → 0: nothing was created.
- Every route is covered the same way by apps/api/src/security/route-guards.test.ts (401 without a token, 403 without the permission).

Automated test: `AT-10 Authorization: a cashier calling user and backup operations directly is refused by the server`

Also covered by: `apps/api/src/security/route-guards.test.ts`, `apps/api/src/admin/users.test.ts`, `apps/api/src/backup/routes.test.ts`, `apps/api/src/auth/authorization.test.ts`

AT-11 Duplicate billing PASS

Test data: Run billing generation and finalization for September 2026 twice more

Expected (spec): No duplicate finalized invoice for same service account/period.

Observed:

- Live invoices for September 2026: 13 before, 13 after two more runs.
- Run 1: generate created 0 drafts; finalize answered HTTP 409 NOTHING_TO_FINALIZE (nothing new to number).
- Run 2: generate created 0 drafts; finalize answered HTTP 409 NOTHING_TO_FINALIZE (nothing new to number).
- No service account has two live invoices for any month.
- Database guard: CREATE UNIQUE INDEX invoices_one_per_period_idx ON public.invoices (service_account_id, period_start) WHERE (status <> 'void'::text).

Automated test: `AT-11 Duplicate billing: generating and finalizing the same month twice creates no second invoice`

Also covered by: `apps/api/src/billing/service.test.ts`, `apps/api/src/billing/schema.test.ts`

AT-12 Backup/restore PASS

Test data: Create a backup, post a ₱100 payment, restore the backup

Expected (spec): Restored database passes integrity check and expected records return.

Observed:

- Backup BK-000097 created and verified (checksums + pg_restore --list).
- Payments: 25 at backup, 26 after the change, 25 after restoring; receipt RCPT-000123 (made after the backup) is gone.
- A safety backup BK-000098 was taken before restoring.
- Integrity check after restore: migrations passed, invoice_paid passed, payment_allocated passed, ledger_balance passed, counters passed, proof_files passed, row_counts passed.
- All sessions were signed out by the restore (old token: HTTP 401).

Automated test: `AT-12 Backup/restore: back up, change data, restore; integrity passes and the records return`

Also covered by: `apps/api/src/backup/restore.test.ts`

All automated tests

Unit tests for the domain and money rules (`packages/shared`), and integration tests for every API service and route against PostgreSQL (`apps/api`), including the acceptance suite above.

Package	Test files	Tests	Passed	Failed
@bcis/shared (domain rules, money, allocation, validation)	17	209	209	0
@bcis/api (services, routes, security, backup, concurrency)	48	551	551	0

Manual and desktop evidence

Desktop workflows are checked by hand against the same rules; screenshots are in `tests/screenshots/`, sample report files in `reports-samples/`, and defects found with their fixes in `docs/bug-log.md`.